

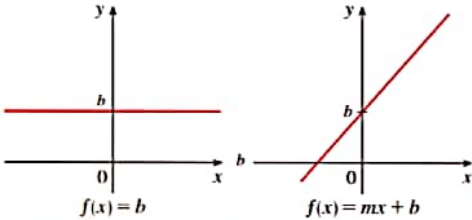
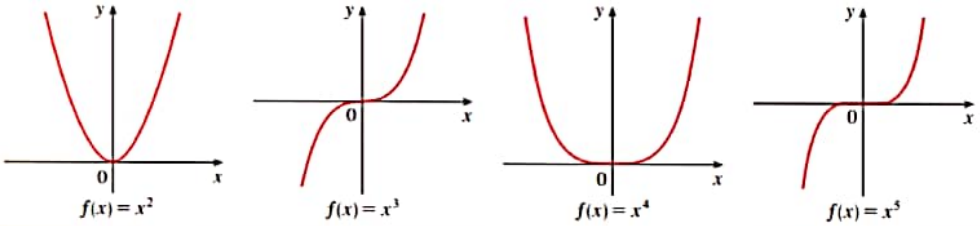
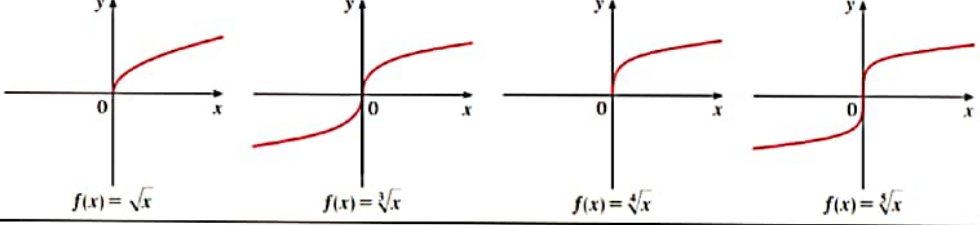
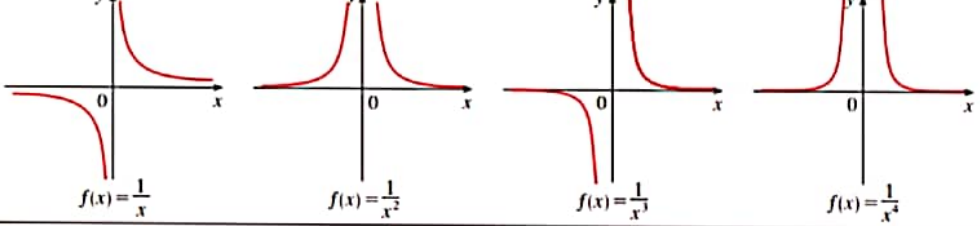
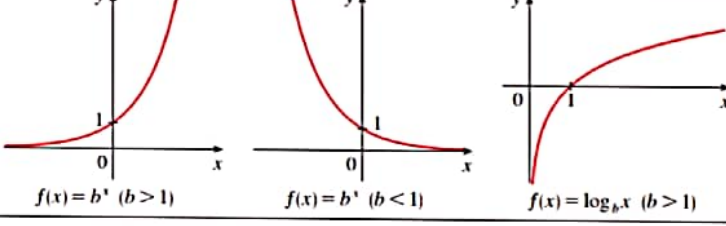
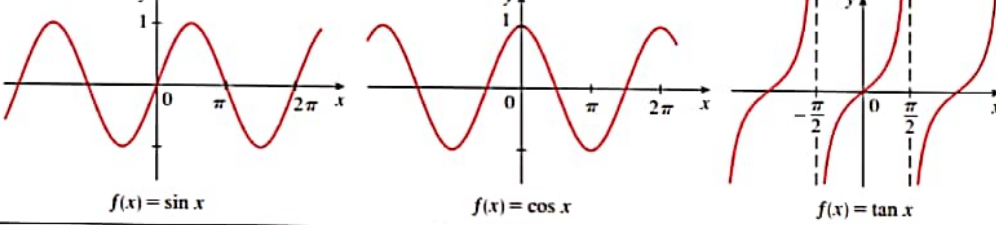
FIGURE 22

- (a) $f(x) = 5^x$ is an exponential function. (The variable x is the exponent.)
 (b) $g(x) = x^5$ is a power function. (The variable x is the base.) We could also consider it to be a polynomial of degree 5.
 (c) $h(x) = \frac{1+x}{1-\sqrt{x}}$ is an algebraic function. (It is not a rational function because the denominator is not a polynomial.)
 (d) $u(t) = 1 - t + 5t^4$ is a polynomial of degree 4.

Table 3 (on the following page) shows a summary of graphs of some families of essential functions that will be used frequently throughout the book.

Copyright 2021 Cengage Learning. All Rights Reserved. May not be copied, scanned, or duplicated, in whole or in part. WCN 02-200-203
 Copyright 2021 Cengage Learning. All Rights Reserved. May not be copied, scanned, or duplicated, in whole or in part. Due to electronic rights, some third party content may be suppressed from the eBook and/or eChapter(s). Editorial review has deemed that any suppressed content does not materially affect the overall learning experience. Cengage Learning reserves the right to remove additional content at any time if subsequent rights restrictions require it.

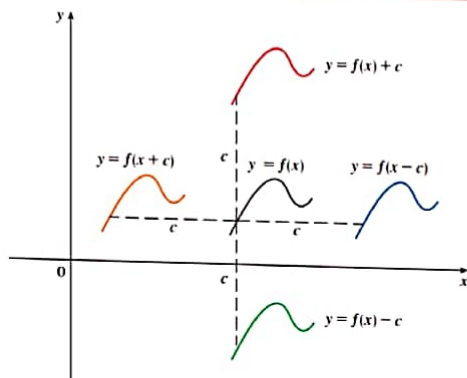
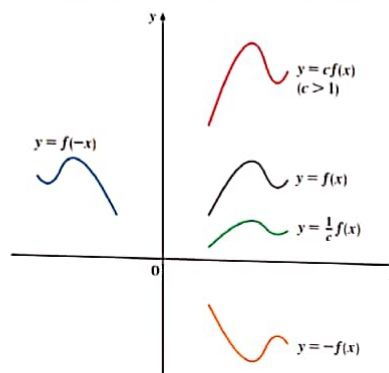
Table 3 Families of Essential Functions and Their Graphs

Linear Functions $f(x) = mx + b$	
Power Functions $f(x) = x^n$	
Root Functions $f(x) = \sqrt[n]{x}$	
Reciprocal Functions $f(x) = \frac{1}{x^n}$	
Exponential and Logarithmic Functions $f(x) = b^x$ $f(x) = \log_b x$	
Trigonometric Functions $f(x) = \sin x$ $f(x) = \cos x$ $f(x) = \tan x$	

Copyright 2021 Cengage Learning. All Rights Reserved. May not be copied, scanned, or duplicated, in whole or in part. WCN 02-200-203
 Copyright 2021 Cengage Learning. All Rights Reserved. May not be copied, scanned, or duplicated, in whole or in part. Due to electronic rights, some third party content may be suppressed from the eBook and/or eChapter(s). Editorial review has deemed that any suppressed content does not materially affect the overall learning experience. Cengage Learning reserves the right to remove additional content at any time if subsequent rights restrictions require it.

Let's first consider **translations** of graphs. If c is a positive number, then the graph of $y = f(x) + c$ is just the graph of $y = f(x)$ shifted upward a distance of c units (because each y -coordinate is increased by the same number c). Likewise, if $g(x) = f(x - c)$, where $c > 0$, then the value of g at x is the same as the value of f at $x - c$ (c units to the left of x). Therefore the graph of $y = f(x - c)$ is just the graph of $y = f(x)$ shifted c units to the right (see Figure 1).

Vertical and Horizontal Shifts Suppose $c > 0$. To obtain the graph of
 $y = f(x) + c$, shift the graph of $y = f(x)$ a distance c units upward
 $y = f(x) - c$, shift the graph of $y = f(x)$ a distance c units downward
 $y = f(x - c)$, shift the graph of $y = f(x)$ a distance c units to the right
 $y = f(x + c)$, shift the graph of $y = f(x)$ a distance c units to the left

FIGURE 1 Translating the graph of f FIGURE 2 Stretching and reflecting the graph of f

Now let's consider the **stretching** and **reflecting** transformations. If $c > 1$, then the graph of $y = cf(x)$ is the graph of $y = f(x)$ stretched by a factor of c in the vertical direction (because each y -coordinate is multiplied by the same number c). The graph of $y = -f(x)$ is the graph of $y = f(x)$ reflected about the x -axis because the point (x, y) is replaced by the point $(x, -y)$. (See Figure 2 and the following chart, where the results of other stretching, shrinking, and reflecting transformations are also given.)

Vertical and Horizontal Stretching and Reflecting Suppose $c > 1$. To obtain the graph of

$y = cf(x)$, stretch the graph of $y = f(x)$ vertically by a factor of c
 $y = (1/c)f(x)$, shrink the graph of $y = f(x)$ vertically by a factor of c
 $y = f(cx)$, shrink the graph of $y = f(x)$ horizontally by a factor of c
 $y = f(x/c)$, stretch the graph of $y = f(x)$ horizontally by a factor of c
 $y = -f(x)$, reflect the graph of $y = f(x)$ about the x -axis
 $y = f(-x)$, reflect the graph of $y = f(x)$ about the y -axis

Copyright 2021 Cengage Learning. All Rights Reserved. May not be copied, scanned, or duplicated, in whole or in part. WCN 02-200-203
 Copyright 2021 Cengage Learning. All Rights Reserved. May not be copied, scanned, or duplicated, in whole or in part. Due to electronic rights, some third party content may be suppressed from the eBook and/or eChapter(s). Editorial review has determined that any suppressed content does not materially affect the overall learning experience. Cengage Learning reserves the right to remove additional content at any time if subsequent rights restrictions require it.

CHAPTER 1 Functions and Models

Figure 3 illustrates these stretching transformations when applied to the cosine function with $c = 2$. For instance, in order to get the graph of $y = 2 \cos x$ we multiply the y -coordinate of each point on the graph of $y = \cos x$ by 2. This means that the graph of $y = \cos x$ gets stretched vertically by a factor of 2.

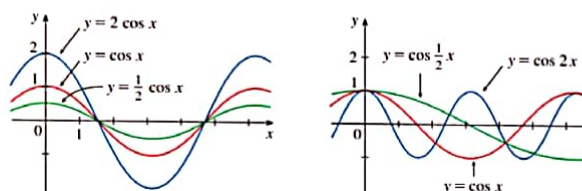


FIGURE 3

EXAMPLE 1 Given the graph of $y = \sqrt{x}$, use transformations to graph $y = \sqrt{x} - 2$, $y = \sqrt{x - 2}$, $y = -\sqrt{x}$, $y = 2\sqrt{x}$, and $y = \sqrt{-x}$.

SOLUTION The graph of the square root function $y = \sqrt{x}$, obtained from Figure 1.2.13(a), is shown in Figure 4(a). In the other parts of the figure we sketch $y = \sqrt{x} - 2$ by shifting 2 units downward, $y = \sqrt{x - 2}$ by shifting 2 units to the right, $y = -\sqrt{x}$ by reflecting about the x -axis, $y = 2\sqrt{x}$ by stretching vertically by a factor of 2, and $y = \sqrt{-x}$ by reflecting about the y -axis.

